125101 - Analytical chemistry 1 for engineers

Tutorial 9 – Potentiometry in Titration

T9-1. Given the following reaction

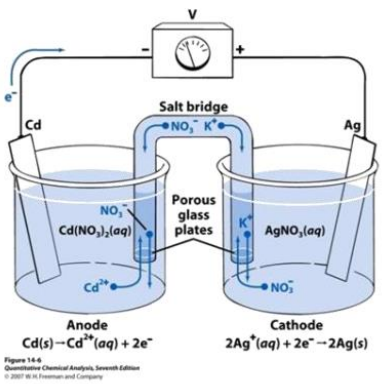
$$Fe(s) \,|\, Fe^{2+} \,||\, Sn^{2+} \,|\, Sn(s)$$

a. Write the reaction of the anode;

b. Write the reaction of the cathode;

c. Write the net reaction.

T9-2. What is the potential E^0 for the following set-up?



$E^0(Cd^{2+}/Cd^0) = -0.402V$

$E^0(Ag^+/Ag^0) = 0.799V$

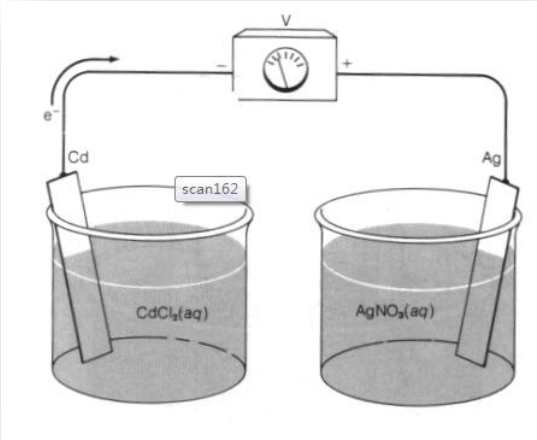
Line diagram for the above cell:

$Cd(s)|CdCl_2(aq)||AgNO_3(aq)|Ag(s)$

9T-3. Try to calculate the cell voltage in the following Figure if the right half-cell contains 0.50 M $AgNO_3$ (aq) and the left half-cell contains 0.010 M $Cd(NO_3)_2$ (aq).

$$E^o(Ag^+|Ag) = 0.799 \text{ V}$$

$$E^o(Cd^{2+}|Cd) = -0.402 \text{ V}$$


V
−
+
e⁻
Cd
Ag
scan162
CdCl₂(aq)
AgNO₃(aq)